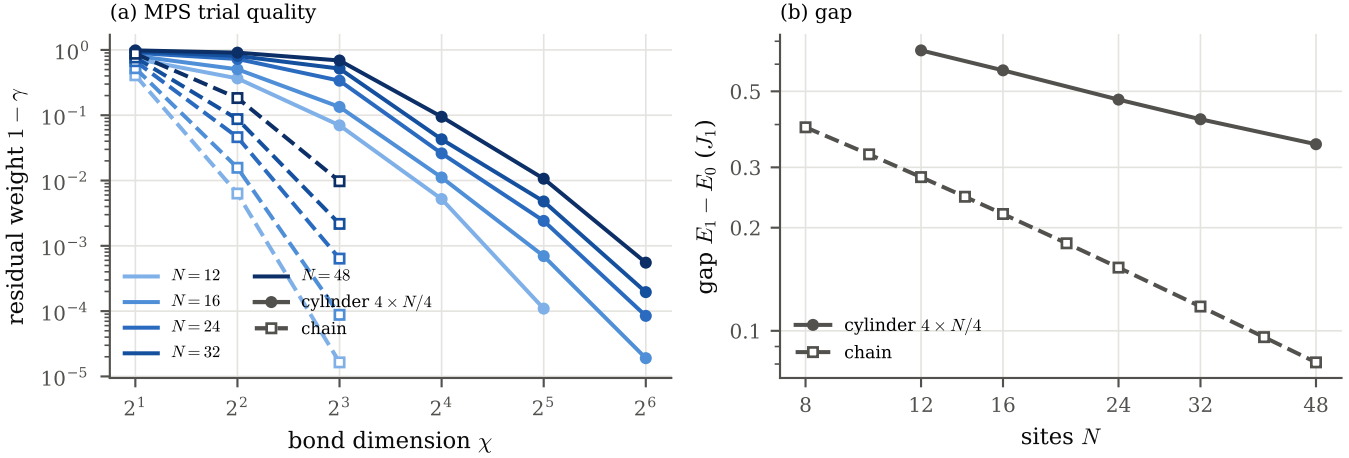
(a) MPS trial quality
residual weight $1-\gamma$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
2^{1}
2^{2}
2^{3}
2^{4}
2^{5}
2^{6}
bond dimension χ
$N=12$
$N=16$
$N=24$
$N=32$
$N=48$
cylinder $4\times N/4$
chain
(b) gap
gap E_1-E_0 (J_1)
0.5
0.3
0.2
0.1
8
12
16
24
32
48
sites N
cylinder $4\times N/4$
chain